

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – Experiments

|                  |                                                                                                                                |               |                                 |
|------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------|
| Course Code:     | DSA03                                                                                                                          | Course Title: | Natural Language Processing     |
| CO Assessed:     | CO3 – Precision                                                                                                                | Assessment:   | Assessment Tool 1 – Experiments |
| Weightage:       | 50% of CIA                                                                                                                     |               |                                 |
| CO5 Statement:   | Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3) |               |                                 |
| Student Name:    | A SHIVA SHANKAR REDDY                                                                                                          | Reg. No.:     | 192425174                       |
| Section / Batch: | CSE-AIML                                                                                                                       | Date:         | 08-08-2026                      |

### Code of Conduct

I A SHIVA SHANKAR REDDY/192425174 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A Shiva Shankar Reddy

### Instructions

- Perform all experiments using Google Colab.
- Create a separate notebook (.ipynb) for the assessment.
- Ensure all code is executable without errors.
- Include appropriate comments explaining the logic used.
- Complete all three experiments in a single Google Colab notebook.
- Execute all cells and display the outputs for the given test cases.
- Save the notebook with the naming convention: RegNo\_Name\_CO3-AT1.ipynb
- Upload the notebook to your GitHub repository.
- Ensure the repository is public or accessible to the instructor.
- Submit the following:
  - GitHub Repository Link in GCR
  - Google Colab Share Link in GCR
  - Screenshots of uploaded coding and output files as printout.
- Duration: 30 minutes.

| Section / Topic                                                                                                                                 | Focus                                                                                                                                               | Experiment Questions |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Context-Free Grammars for English Syntax, Context-Free Rules and Trees, Sentence-Level Constructions, Parsing, Top-Down Parsing, Earley Parsing | Design a Python program to validate sentences using CFG production rules and construct parse tree representations for valid sentences.              | Q1                   |
| Agreement, Feature Structures, Sub Categorization                                                                                               | Design a Python program to verify subject-verb agreement and validate grammatical constraints using feature structures and subcategorization rules. | Q2                   |
| Probabilistic Context-Free Grammars (PCFGs)                                                                                                     | Design a Python program to parse sentences and compute sentence probabilities using probabilistic grammar production rules.                         | Q3                   |

Experiment 1: Context-Free Grammar Rule Validation and Parse Tree Construction

Experiment Description

Design a Python program to validate whether a sentence follows the given Context-Free Grammar (CFG) rules and generate its parse tree representation.

Grammar:

```
S → NP VP
NP → Det N
VP → V NP

Det → the | a
N  → student | teacher | book
V   → reads | likes
```

Task

- 1. Accept a sentence as input.
- 2. Verify whether it conforms to the CFG.
- 3. Construct and return the parse tree.
- 4. Return "Invalid Sentence" if parsing fails.

Function Prototype

```
def parse_cfg(sentence):
    pass
```

Test Cases

| Input                    | Expected Output  |
|--------------------------|------------------|
| the student reads a      | Valid Parse Tree |
| book a teacher likes the | Valid Parse Tree |
| book                     |                  |
| student reads book       | Invalid Sentence |
| the book likes a teacher | Valid Parse Tree |
| reads the student book   | Invalid Sentence |

Aim:

To design a Python program that validates whether a sentence follows the given Context-Free Grammar (CFG) rules and generates its parse tree representation.

Problem Statement:

Design a Python program to:

- 1. Accept a sentence as input.
- 2. Verify whether it conforms to the given CFG.
- 3. Construct and return the parse tree.
- 4. Return "Invalid Sentence" if parsing fails.

## Grammar Rules:

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$VP \rightarrow V NP$

$Det \rightarrow the \mid a$

$N \rightarrow student \mid teacher \mid book$

$V \rightarrow reads \mid likes$

## Algorithm:

1. Define the CFG grammar rules.
2. Accept a sentence from the user.
3. Tokenize the sentence into words.
4. Parse the sentence using CFG.
5. If parsing succeeds:

Display "Valid Sentence".

Generate and display the parse tree.

6. Otherwise:

1. Display "Invalid Sentence".

## Python Code:

```
import nltk from nltk import CFG
from nltk.parse import ChartParser
```

```
grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V NP
Det -> 'the' | 'a'
N -> 'student' | 'teacher' | 'book'
V -> 'reads' | 'likes' """) parser
= ChartParser(grammar)
```

```
def parse_cfg(sentence): words =
    sentence.lower().split()
```

```
try: trees =
    list(parser.parse(words))
```

```
    if trees:
        print("Valid Sentence")
        for tree in trees:
            print(tree)
    else: print("Invalid
        Sentence")
```

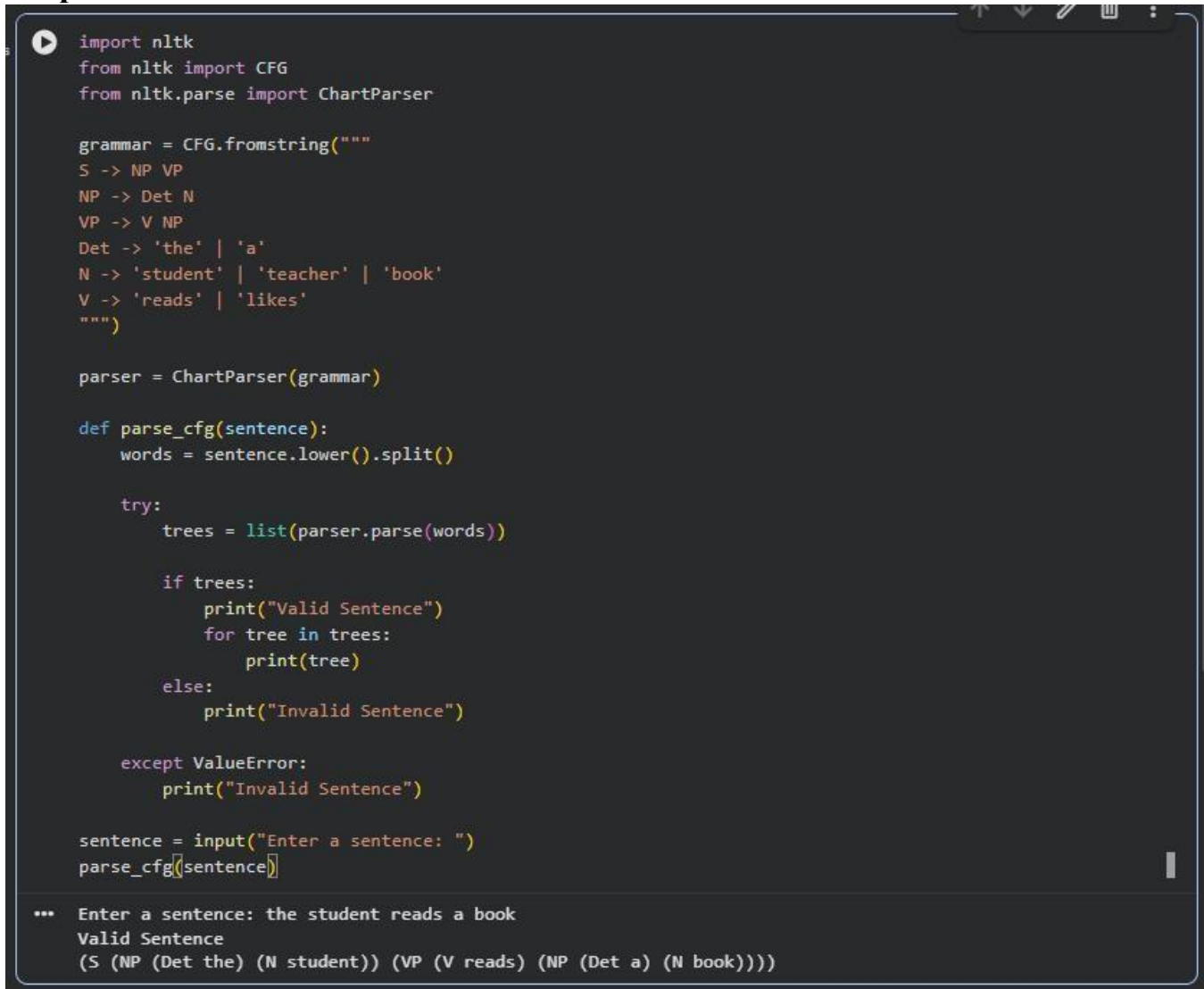
```
except ValueError:
```

```
print("Invalid Sentence")
```

```
sentence = input("Enter a sentence: ")
```

```
parse_cfg(sentence)
```

## Output Screenshot:



```
import nltk
from nltk import CFG
from nltk.parse import ChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V NP
Det -> 'the' | 'a'
N -> 'student' | 'teacher' | 'book'
V -> 'reads' | 'likes'
""")

parser = ChartParser(grammar)

def parse_cfg(sentence):
    words = sentence.lower().split()

    try:
        trees = list(parser.parse(words))

        if trees:
            print("Valid Sentence")
            for tree in trees:
                print(tree)
        else:
            print("Invalid Sentence")

    except ValueError:
        print("Invalid Sentence")

sentence = input("Enter a sentence: ")
parse_cfg(sentence)
```

\*\*\* Enter a sentence: the student reads a book  
Valid Sentence  
(S (NP (Det the) (N student)) (VP (V reads) (NP (Det a) (N book))))

## Test Cases and Output:

**Test Case 1 Input:** the  
student reads a book **Output**

Valid Sentence

(S  
  (NP (Det the) (N student))  
  (VP (V reads)  
      (NP (Det a) (N book))))

### **Test Case 2 Input a**

teacher likes the book

### **Output**

Valid Sentence

```
(S
  (NP (Det a) (N teacher))
  (VP (V likes)
    (NP (Det the) (N book))))
```

### **Test Case 3 Input**

student reads book

### **Output**

Invalid Sentence

### **Test Case 4 Input the**

book likes a teacher

### **Output**

Valid Sentence

```
(S
  (NP (Det the) (N book))
  (VP (V likes)
    (NP (Det a) (N teacher))))
```

### **Test Case 5 Input**

reads the student book

### **Output**

Invalid Sentence

These outputs correspond to the test cases specified in the assessment.

## **Result Analysis:**

- Sentences following the CFG structure  $S \rightarrow NP VP$  are successfully parsed.
- Parse trees clearly represent the grammatical structure.
- Sentences violating CFG rules are correctly identified as invalid.
- The program effectively validates English sentence syntax using Context-Free Grammar.

## Result:

The Python program successfully validates sentences using the given CFG rules and generates parse tree representations for valid sentences while rejecting invalid sentences. This demonstrates the application of CFG and parsing techniques in Natural Language Processing.

## Experiment 2: Agreement and Feature Structure Checking

### Experiment Description

Design a Python program that verifies subject–verb agreement using feature structures containing Number and Person attributes. **Feature Rules** he → singular, third person she → singular, third person it → singular, third person they → plural

runs → singular verb  
writes → singular verb

run → plural verb  
write → plural verb **Task**

1. Accept Subject and Verb.
2. Compare feature structures.
3. Return True if agreement holds.
4. Return False otherwise.

### Function Prototype

```
def check_agreement(subject, verb):  
    pass
```

### Test Cases

#### Subject Verb Expected

he runs True he run  
False they run True  
they writes False she  
writes True

## Aim

To design a Python program that verifies subject–verb agreement using feature structures containing Number and Person attributes and validates grammatical constraints.

## Problem Statement:

Design a Python program that:

1. Accepts a Subject and a Verb.
2. Compares their feature structures.
3. Returns True if agreement holds.
4. Returns False otherwise.

## Feature Rules

| Subject      | Feature         |
|--------------|-----------------|
| Structure he | singular, third |
| person she   | singular, third |
| person it    | singular, third |
| person they  | plural          |

| Verb        | Feature     | Structure |
|-------------|-------------|-----------|
| runs        | singular    | verb      |
| writes      | singular    | verb run  |
| plural verb | write       |           |
|             | plural verb |           |

### Algorithm:

1. Define feature structures for subjects and verbs.
2. Accept subject and verb as input.
3. Extract Number and Person features.
4. Compare subject features with verb features.
5. If features match:
  1. Return True
6. Otherwise:
  1. Return False
7. Display the result.

### Python Code: def

check\_agreement(subject, verb):

```
subject_features = {
    "he": ("singular", "third"),
    "she": ("singular", "third"),
    "it": ("singular", "third"),
    "they": ("plural", "third")
}
```

```
verb_features = {
    "runs": "singular",
    "writes": "singular",
    "run": "plural",
    "write": "plural" } if subject not in subject_features or
```

```
verb not in verb_features: return False
```

```
subject_number =  
subject_features[subject][0] verb_number =  
verb_features[verb] return subject_number ==  
verb_number
```

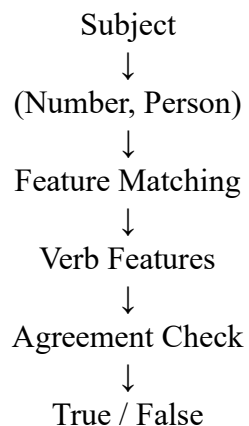
```
subject = input("Enter Subject: ").lower() verb = input("Enter Verb: ").lower()  
print(check_agreement(subject, verb))
```

## Output Screenshot:

```
def check_agreement(subject, verb):  
  
    subject_features = {  
        "he": ("singular", "third"),  
        "she": ("singular", "third"),  
        "it": ("singular", "third"),  
        "they": ("plural", "third")  
    }  
  
    verb_features = {  
        "runs": "singular",  
        "writes": "singular",  
        "run": "plural",  
        "write": "plural"  
    }  
  
    if subject not in subject_features or verb not in verb_features:  
        return False  
  
    subject_number = subject_features[subject][0]  
    verb_number = verb_features[verb]  
  
    return subject_number == verb_number  
  
subject = input("Enter Subject: ").lower()  
verb = input("Enter Verb: ").lower()  
  
print(check_agreement(subject, verb))  
  
*** Enter Subject: he  
    Enter Verb: runs  
    True
```

## Experimental Design

### Feature Structure Representation



## **Test Cases and Outputs**

### **Test Case 1**

**Input :** he runs

**Output**

True

### **Test Case 2**

**Input :** he run

**Output**

False

### **Test Case 3**

**Input :** they run

**Output**

True

### **Test Case 4**

**Input :** they writes

**Output**

False

### **Test Case 5**

**Input :** she writes

**Output:**

True

These outputs match the expected test cases provided in the assessment document.

## **Result Analysis:**

- Singular subjects (he, she, it) correctly match singular verbs (runs, writes).
- Plural subjects (they) correctly match plural verbs (run, write).
- Mismatched Number features are identified and rejected.
- The feature structure approach effectively validates grammatical constraints and subject–verb agreement.

**Result:**

The Python program successfully verifies subject–verb agreement using feature structures based on Number and Person attributes. It correctly returns True for grammatically valid combinations and False for invalid combinations, demonstrating the application of feature structures and subcategorization rules in Natural Language Processing.

### Experiment 3: Probabilistic Context-Free Grammar (PCFG) Parsing

#### Experiment Description

Implement a Python program to compute the probability of a sentence using the given PCFG.

#### Grammar

$S \rightarrow NP VP$  [1.0]

$NP \rightarrow John$  [0.6]

$NP \rightarrow Mary$  [0.4]

$VP \rightarrow runs$  [0.5]

$VP \rightarrow walks$  [0.5]

#### Probability Formula

$P(\text{Sentence})$

=

$P(S \rightarrow NP VP)$

$\times P(NP)$

$\times P(VP)$

#### Task

1. Accept a two-word sentence.
2. Parse using the PCFG.
3. Compute sentence probability.
4. Return 0 if sentence cannot be generated.

#### Function Prototype

```
def sentence_probability(sentence):  
    pass
```

#### Test Cases

| Input Sentence Expected Probability |      |
|-------------------------------------|------|
| John runs                           | 0.30 |
| John walks                          | 0.30 |
| Mary runs                           | 0.20 |
| Mary walks                          | 0.20 |
| Peter runs                          | 0.00 |

#### Aim:

To design a Python program that parses sentences and computes sentence probabilities using Probabilistic Context-Free Grammar (PCFG) production rules.

#### Problem Statement:

Design a Python program to:

1. Accept a two-word sentence.

2. Parse the sentence using the given PCFG.
3. Compute the sentence probability.
4. Return **0** if the sentence cannot be generated by the grammar.

### PCFG Grammar:

$S \rightarrow NP VP [1.0]$

$NP \rightarrow John [0.6]$

$NP \rightarrow Mary [0.4]$

$VP \rightarrow runs [0.5]$

$VP \rightarrow walks [0.5]$

Probability Formula

$P(\text{Sentence})$

$= P(S \rightarrow NP VP)$

$\times P(NP)$

$\times P(VP)$

### Algorithm

1. Define the probabilities of NP and VP productions.
2. Accept a sentence from the user.
3. Split the sentence into NP and VP.
4. Check whether the words exist in the grammar.
5. Compute the probability using:

**$P(\text{Sentence}) = P(S \rightarrow NP VP) \times P(NP) \times P(VP)$**

1. If any word is not present in the grammar:

1. Return 0.00

2. Display the final probability.

### Python Code: def

sentence\_probability(sentence):

```
np_probs = {
    "John": 0.6,
    "Mary": 0.4
}
```

```
vp_probs = {
    "runs": 0.5,
    "walks": 0.5 } words
```

```
= sentence.split()
```

```

if len(words) != 2:
    return 0.0 np_word,
    vp_word = words

if np_word not in np_probs:
    return 0.0

if vp_word not in vp_probs: return 0.0 probability = 1.0 *
    np_probs[np_word] * vp_probs[vp_word] return probability

```

sentence = input("Enter Sentence: ")print("Sentence Probability =", sentence\_probability(sentence))

## Output Screenshot :

```

def sentence_probability(sentence):

    np_probs = {
        "John": 0.6,
        "Mary": 0.4
    }

    vp_probs = {
        "runs": 0.5,
        "walks": 0.5
    }

    words = sentence.split()

    if len(words) != 2:
        return 0.0

    np_word, vp_word = words

    if np_word not in np_probs:
        return 0.0

    if vp_word not in vp_probs:
        return 0.0

    probability = 1.0 * np_probs[np_word] * vp_probs[vp_word]

    return probability

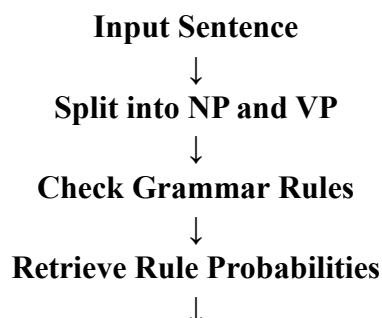
sentence = input("Enter Sentence: ")
print("Sentence Probability =", sentence_probability(sentence))

... Enter Sentence: John runs
Sentence Probability = 0.3

```

## Experimental Design:

PCFG Parsing Process



## Apply PCFG Formula



## Sentence Probability

### Test Cases and Outputs

#### Test Case 1

**Input :** John runs

#### Calculation

$$\begin{aligned} P &= 1.0 \times 0.6 \times 0.5 \\ &= 0.30 \end{aligned}$$

#### Output

Sentence Probability = 0.30

#### Test Case 2

**Input :** John walks

#### Calculation

$$\begin{aligned} P &= 1.0 \times 0.6 \times 0.5 \\ &= 0.30 \end{aligned}$$

#### Output

Sentence Probability = 0.30

#### Test Case 3

**Input :** Mary runs

#### Calculation

$$\begin{aligned} P &= 1.0 \times 0.4 \times 0.5 \\ &= 0.20 \end{aligned}$$

#### Output

Sentence Probability = 0.20

#### Test Case 4

**Input :** Mary walks

#### Calculation

$$\begin{aligned} P &= 1.0 \times 0.4 \times 0.5 \\ &= 0.20 \end{aligned}$$

## Output

Sentence Probability = 0.20

## Test Case 5

**Input :** Peter runs

## Calculation

Peter is not present in NP rules

## Output

Sentence Probability = 0.00

These outputs match the expected results given in the assessment document.

## Result Analysis:

- The PCFG parser successfully computes sentence probabilities using grammar production rules.
- Sentences generated by the grammar receive valid probabilities.
- Sentences containing words outside the grammar receive a probability of **0.00**.
- Higher probability values indicate more likely sentence constructions according to the grammar.
- The approach demonstrates how probabilistic grammars are used in NLP parsing and language modeling.

## Result:

The Python program successfully parses valid sentences and computes their probabilities using the given Probabilistic Context-Free Grammar. Invalid sentences that cannot be generated by the grammar are correctly assigned a probability of **0.00**. This demonstrates the practical application of PCFGs for probabilistic sentence parsing and syntactic analysis in Natural Language Processing.

## Rubrics for Calculation

| Criteria                  | Weightage | Level 4 – Exemplary                                                                           | Level 3 – Proficient                                        | Level 2 – Developing                                | Level 1 – Beginning                              |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Understanding of Concepts | 25        | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design       | 30        | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation           |
| Use of Tools              | 20        | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools                |

|                     |    |                                                                            |                                              |                                      |                                       |
|---------------------|----|----------------------------------------------------------------------------|----------------------------------------------|--------------------------------------|---------------------------------------|
| Analysis of Results | 15 | Insightful analysis with accurate interpretation and validation of results | Good analysis with reasonable interpretation | Basic analysis with limited insights | Poor or incorrect analysis of results |
|---------------------|----|----------------------------------------------------------------------------|----------------------------------------------|--------------------------------------|---------------------------------------|

|               |    |                                                                       |                                                |                                          |                                  |
|---------------|----|-----------------------------------------------------------------------|------------------------------------------------|------------------------------------------|----------------------------------|
| Documentation | 10 | Clear, well-structured documentation with diagrams, code, and results | Organized documentation with necessary details | Basic documentation with missing details | Poor or incomplete documentation |
|---------------|----|-----------------------------------------------------------------------|------------------------------------------------|------------------------------------------|----------------------------------|

| Q. No. / Criteria Level | Understanding of Concepts | Experimental Design | Use of Tools | Analysis of Results | Documentation | Sub. Total | Total % |
|-------------------------|---------------------------|---------------------|--------------|---------------------|---------------|------------|---------|
| 1                       |                           |                     |              |                     |               |            |         |
| 2                       |                           |                     |              |                     |               |            |         |
| 3                       |                           |                     |              |                     |               |            |         |

| Faculty In-charge |  | Course Coordinator |  | Program Director |  |
|-------------------|--|--------------------|--|------------------|--|
|                   |  |                    |  |                  |  |
| Signature: _____  |  | Signature: _____   |  | Signature: _____ |  |
| Date: _____       |  | Date: _____        |  | Date: _____      |  |